

WASP-52b

R-1409

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_211012 · created 2026-08-01T05:00:04+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459499.7128 +/- 0.0065 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.213 +/- 0.061
Transit depth (Rp/Rs)^2	4.55 +/- 2.62 [%]
Orbital Inclination (inc)	84.52 +/- 2.79
Airmass coefficient 1 (a1)	5.83 +/- 3.08
Airmass coefficient 2 (a2)	-1.51 +/- 1.3
Scatter in the residuals of the lightcurve fit is	56.15 %
Best Comparison Star	None
Optimal Aperture	2.31
Optimal Annulus	5.38
Transit Duration (day)	0.067 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.213 ± 0.061 vs 0.1646 (deviation 0.79σ)
Duration fit vs published	0.067 d vs 0.0758 d (deviation 0.34σ)
Mid-time O–C	-7.6 min
Depth SNR	3.5
Residual scatter	56.15 %

Light curve

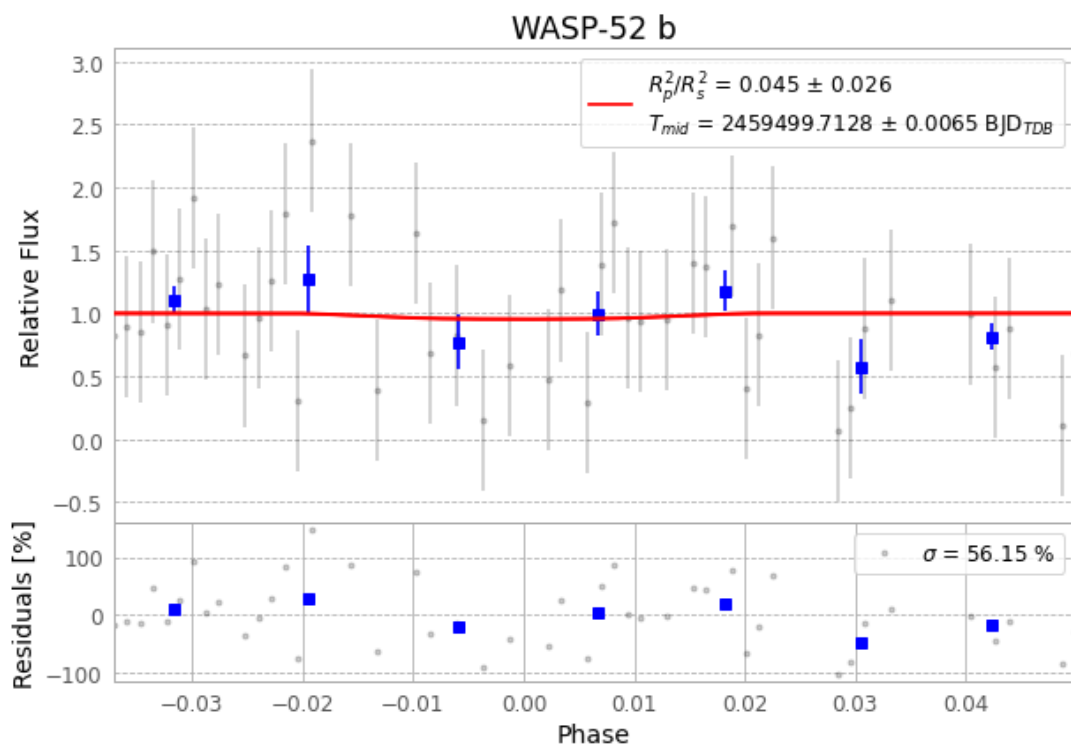


Figure 1 — FinalLightCurve_WASP-52 b_2021-10-11.png (SHA-256 5844a8818e77926a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [398, 88], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4e2ee6e758896cd7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-52211012031812.FITS → WASP-52211012070612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-211012.log (7f2f1109c650...), FinalLightCurve_WASP-52 b_2021-10-11.png (5844a8818e77...), FinalLightCurve_WASP-52 b_2021-10-11.csv (e177d999b04b...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 05:00Z.